

Project I: Software Implementation and Unit Testing

Assignment #2

Overview

This assignment should be done **individually**. Do your own work and **do not share** your work with others. Sharing work is an academic offense and is subject to penalty. Be aware that source code and documents are automatically checked by eConestoga against every other student's work in the course. Academic offenses will be reported to the College Registrar.

You are to create a series of tests for a C application that allows a user to work with and modify a rectangle. You will use GitHub as source control for this assignment.

What to do

Step 1: Setup GitHub and Visual Studio

Download the CSCN71020_A2 solution from eConestoga. Log into your [GitHub.com](https://github.com) account and create a repository for the base code.

Open Visual Studio 2019. In the Get Started window, select Clone a repository.

For the Repository Location, use the URL just created in your personal GitHub account.

NOTE: If you need help performing the above tasks, refer to the instructions in Assignment #1.

You should be able to run the application now. Play around with it to see how it operates. The requirements of this program state:

- It only needs to remember one rectangle at a time
- Rectangle dimensions are to be measured using integers
- Both the length and width of the rectangle is to be limited to the range of values 1 to 99, inclusive.

In Visual Studio, add a **Native Unit Test Project** to the given solution. **Remember the naming conventions** for the Test Project, the Test Class, and the Test Methods. You will need to setup this Test Project to interact with the application project, as described in class.

Make regular commits to your personal repo during the course of this assignment, whenever it makes sense for your situation.

Step 2: Designing and Writing Unit Tests

Task 1

Write one unit tests for each of the following functions:

```
int getPerimeter(int *length, int *width);  
int getArea(int *length, int *width);
```

The requirements of these functions is to return the perimeter and area for a rectangle with the given length and width dimensions. Since input is validated elsewhere in the program, these functions do not have a limit on the range of integers they allow. Each unit test is marked, based on the proper setup, execution, assert, naming and test choice. Cut and paste your test code in the box below:

Task 2

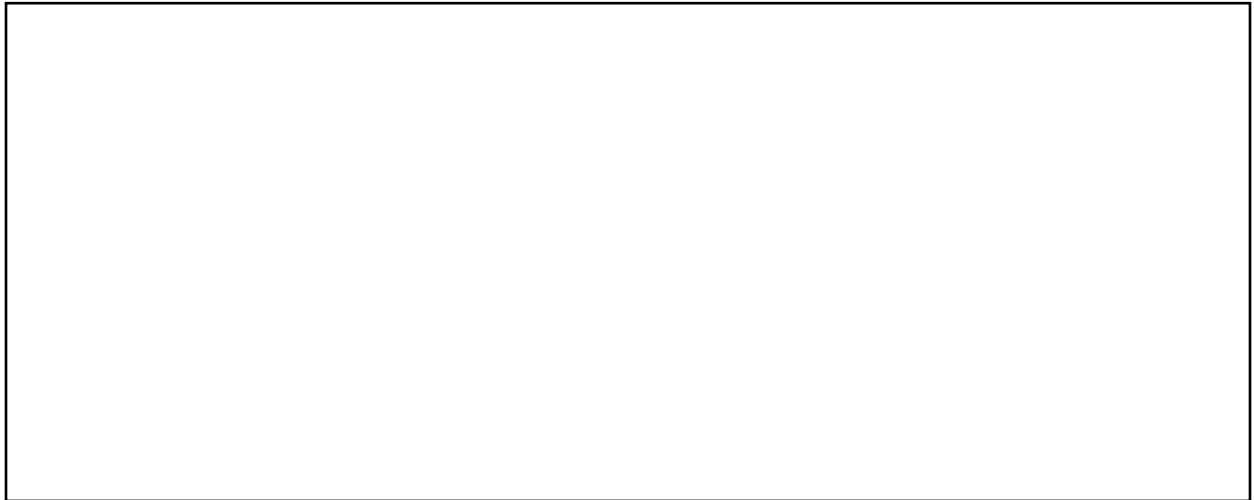
Write three unit tests for each of the following functions:

```
void setLength(int input, int *length);  
void setWidth(int input, int *width);
```

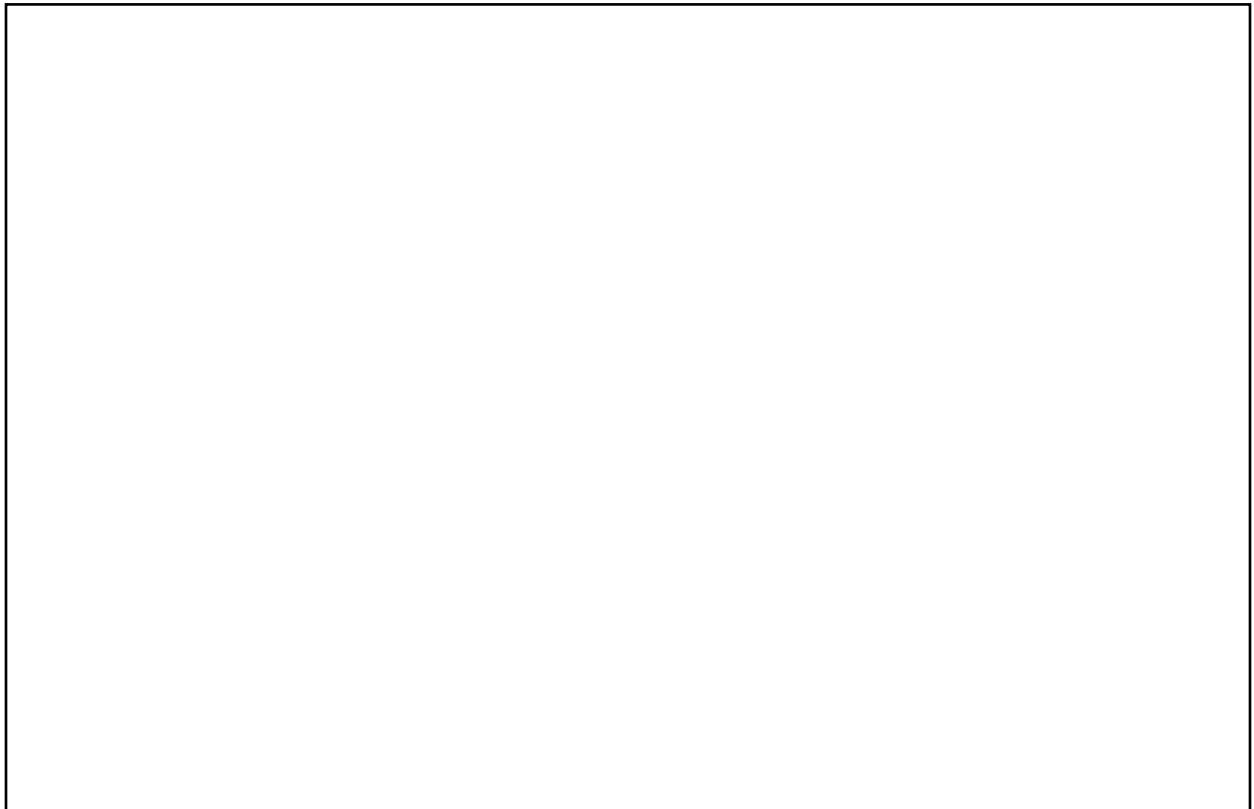
The requirements of these functions is to set the length or width of a rectangle to be any integer from 1 to 99, inclusive. Each unit test is marked based on the proper setup, execution, assert, naming and test choice. Each unit test for the same function should attempt to test some different aspect of the function. Deductions will be given for tests that are testing similar inputs. For different inputs, think of usual, unusual, special or edge cases. Cut and paste your test code in the box below:

Task 3

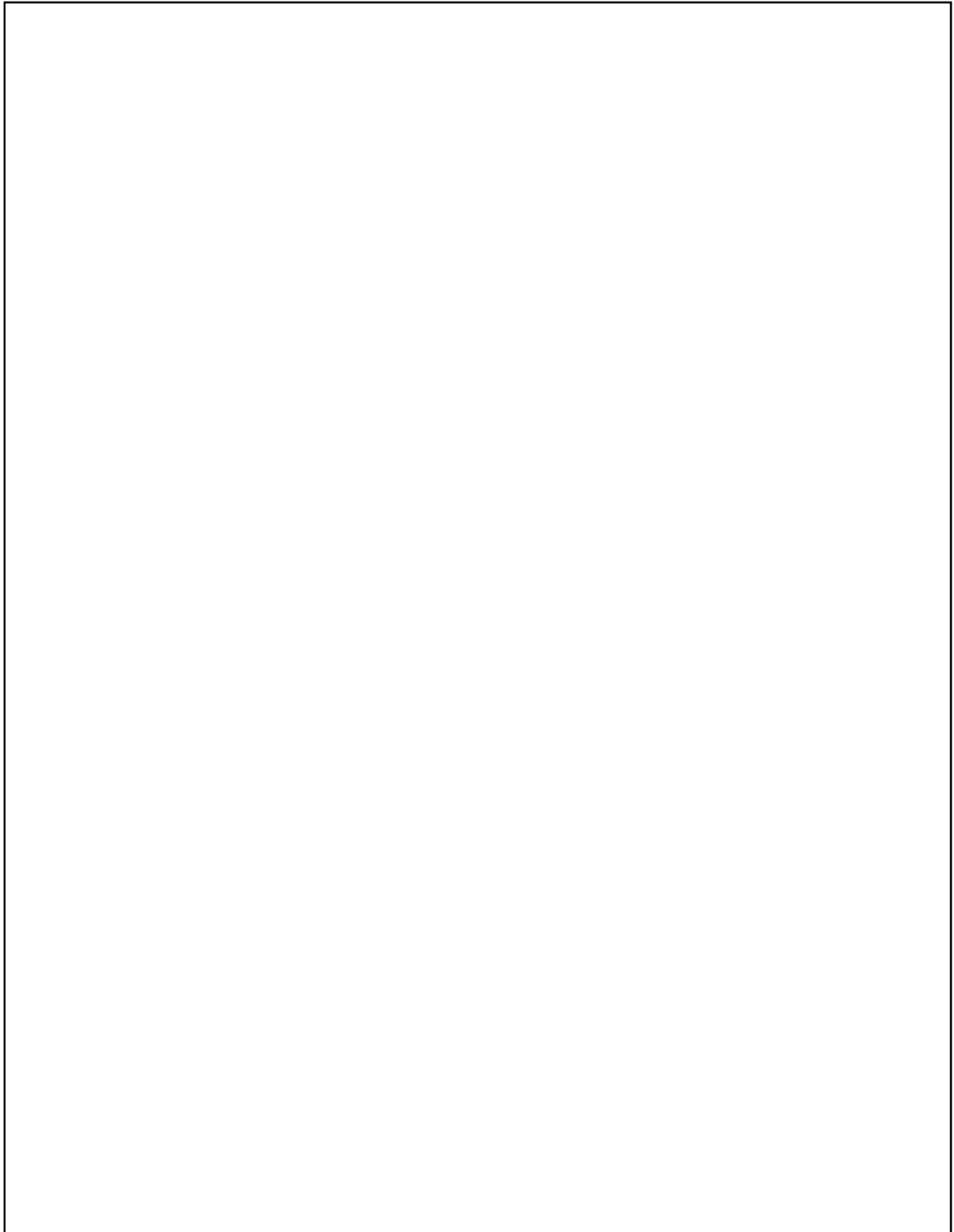
Run your test suite of 8 unit tests against the application code. Include a screenshot of this run in the box below.

A large, empty rectangular box with a black border, intended for a screenshot of the test run results.

If any test fails, briefly explain why the test failed (1 to 2 sentences) and fix either the application or test code. If you had to fix any tests or the application, include the changed code with your explanation in the box below. Marks here are based on what issues were discovered and fixed, and how logical your explanations are if any are or are not found.

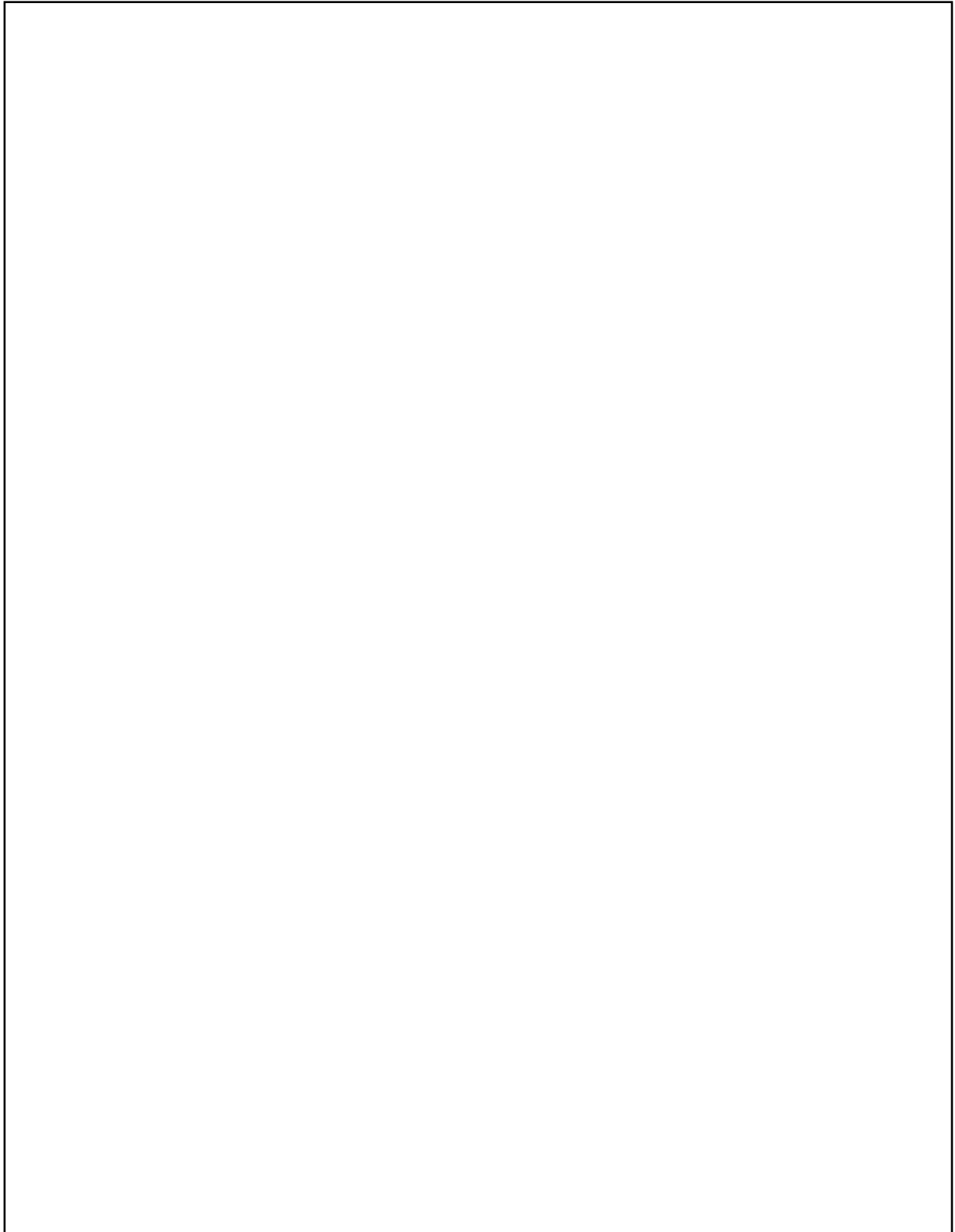
A large, empty rectangular box with a black border, intended for a detailed explanation of any test failures and the corresponding fixes.

If you had any failing tests, after fixing the issue include another screenshot of all the tests passing in the box below:



Task 4

Push your local repo to your GitHub repository. Include a screenshot of your commits page in the box below:



Rubric

See eConestoga.

What to Hand In

Upload the following files to eConestoga using the Assignment #2 link

- A copy of this PDF form properly filled out
- A single compressed (.zip format) archive file containing **the entire Solution folder** of your source code (so I can run it as required).

Do not reformat this form in any way. Just fill it out, save it and upload it. Modifying this form will result in a 10% penalty.